

Allan Vikiru

Machine Learning | Edge Computing | Internet of Things

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[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

Career Profile

- A junior researcher of Computer Science interested in PhD opportunities that explore the intersection of internet-of-things and machine learning to develop remote sensing or edge analytics applications.
- Knowledgeable in underlying technologies of edge AI, with Python (scikit-learn), R, Arduino and ESP32 as key technologies and tools.
- Demonstrates progressive research experience with engagement in 4 research publications and 3 projects that involved development of IoT architectures and machine learning models.

Skills and Competencies

Programming Languages

C++, JavaScript, Python (scikit-learn, TensorFlow, Django), R, MATLAB

Development Platforms and Software

AWS, AutoCAD, Arduino, ESP32, Git, Microsoft Azure IoT, QGIS, Raspberry Pi, Streamlit, Thingspeak, Wokwi

Professional Skills

Technical Writing, Commercial and Academic Presentations, Project Management, Stakeholder Engagement

Education

MSc. Information Technology (Business Intelligence & Data Analytics Option) [graduated June 2024]

Strathmore University - Nairobi, Kenya

GPA – 3.64/4 (A)

BSc. Informatics & Computer Science [graduated June 2021]

Strathmore University - Nairobi, Kenya

GPA – 3.5/4 (A)

Publications

1. **Vikiru, A.**, Simiyu, D., Kahoro, S., Muti, Z., Kamau, S., Macharia, A., Zemp, D.C., Muchiri, H.: Kenya Forest Sounds Dataset. [Dataset], (2026). <https://doi.org/10.5281/zenodo.21443074>.
2. **Vikiru A.** & Khakata E. A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters. [Conference Paper], (in press) *11th International Congress on Information and Communication Technology*.
3. **Vikiru, A.**: A Predictive Model for Determining Vegetable-derived Macronutrients for a Diabetic Patient [Thesis], (2024). <https://hdl.handle.net/11071/16545>.
4. Simiyu, D., Muchiri, H., **Vikiru, A.**, Butime, J., Muti, Z.: A Forest Acoustics – Temporal Frequency Convolution Neural Network Model for Detecting Illegal Logging Activities in Forest. [Conference Paper] In: 2024 5th International Conference on Smart Sensors and Application (ICSSA). pp. 1–6 (2024). <https://doi.org/10.1109/ICSSA62312.2024.10788639>.
5. Ayankoso, S., Wang, Z., Shi, D., Yang, W., **Vikiru, A.**, Kamau, S., Muchiri, H., Gu, F.: Development of Long-Range, Low-Powered and Smart IoT Device for Detecting Illegal Logging in Forests. [Conference Paper] *Journal of Dynamics*,

Monitoring and Diagnostics. (2024). <https://doi.org/10.37965/jdmd.2024.550>.

6. Simiyu, D., **Vikiru, A.**, Muchiri, H., Gu, F., Butime, J.: A Chainsaw-Sound Recognition Model for Detecting Illegal Logging Activities in Forests. [Conference Paper] In: Ball, A.D., Ouyang, H., Sinha, J.K., and Wang, Z. (eds.) Proceedings of the UNified Conference of DAMAS, IncoME and TEPEN Conferences (UNified 2023). pp. 797–806. Springer Nature Switzerland, Cham (2024). https://doi.org/10.1007/978-3-031-49421-5_65.
7. **Vikiru, A.**: Market Basket Analysis using Apriori Algorithm [Web Article] (2023) <https://allanvikiru.github.io/MarketBasketAnalysis/>.
8. **Vikiru, A.**, Muiruri, M., Ateya, I.: An Overview on Cloud Distributed Databases for Business Environments [Preprint] (2023). <https://doi.org/10.48550/arXiv.2301.10673>.

Research Experience

Curbing Illegal Logging Practices using Sound Based Detection Techniques [July 2023 – December 2024]

- Supported by Climate Change AI Grants 2023, this project involved creating an edge AI architecture in forested environments to determine logging sounds by chainsaw and axe as a detective measure for illegal logging activities. This research project was conducted under supervision of Dr Henry Muchiri (Strathmore University).
- My roles in this project were in the IoT prototyping in the design and implementation of edge node device and complementary LoRaWAN architecture; collection, analysis and publication of raw sound data; coordinating project meetings and field visits; proposal writing and project documentation.
- A key result from this project was the real-world implementation and testing of a LoRaWAN architecture with on-edge processing capabilities of acoustic classification.

A Predictive Model for Determining Vegetable Derived Macronutrients for a Diabetic Patient [May 2023 – June 2024]

- Supported by the National Research Fund (Kenya), this work featured the development of a regression-based geospatial map and recommendation-based tool that can be used to determine suitable plant vegetables for diabetic patients in Kenya based on the distribution of soil macronutrients. The project served as the thesis topic in fulfillment of the MSc. Information Technology degree under supervision of Prof. Vincent Omwenga (Strathmore University).
- My roles in this project were in the geospatial modelling of georeferenced soil maps; development of an ensemble clustering-classification recommendation model; proposal writing and project documentation.
- A key result from this project was implementing a data fusion technique through a pipeline that involves modelling of geospatial data.

Detecting the Risk of Fireground Injuries among Firefighters using IoT [April 2020 – September 2021]

- This study looked into developing a wearable device that firefighters can wear in the line of duty, to sense their vitals such as heart rate and blood oxygen and environmental parameters such as atmospheric temperature to determine the risk of sustaining fireground injuries. The project served as the final year project of the BSc. Informatics & Computer Science degree under supervision of Dr. Esther Khakata (Strathmore University).
- My roles in this project were in the IoT prototyping in the design and implementation of the node wearable device and complementary GSM architecture; analysis of node-collected data; and developing a web application prototype for presenting real-time results; proposal writing and project documentation
- A key result from this project was the implementation of a sensing-to-intelligence IoT architecture following human-computer interaction principles. In the early stages of dissemination, this project was presented in a national IoT and AI prototyping contest and received recognition as a Top 10 finalist project.

Work Experience

Teaching Fellow – School of Computing and Engineering Sciences, Strathmore University [July 2024 – Present]

- Teaching and assessing undergraduate students in courses on Artificial Intelligence, Embedded Systems & IoT, Information

Security, Data Governance and Cloud Computing.

- Supervising undergraduate students on system development projects with deep learning and Internet of Things as key technologies as well as projects that apply the challenge-driven education methodology.

Research Coordinator – SCES Makerspace Lab, Strathmore University [July 2023 – Present]

- Continually engaging a team of 6 faculty researchers and at least 10 undergraduate and graduate students in developing projects in deep learning and robotics.
- Co-lead of the **Conservation Technology Research Hub**, where I coordinate data collection and analysis activities, model and device prototyping, grant proposal writing and project dissemination among an internal team of 7 members and 3 external partners.

Graduate Assistant – School of Computing & Engineering Sciences, Strathmore University [March 2022 – June 2024]

- Supported senior faculty in teaching and research activities in facilitating laboratory sessions, assessing coursework and developing research proposals.

Software Developer Intern – IoT & Wireless Networks Lab, @iLabAfrica [January 2022 - March 2022]

- Designed and implemented breadboard models for node devices, and a monitoring dashboard in Vue.js and Node.js with a Keycloak authentication system for a research project that involved LoRaWAN communication.

Certifications & Licenses

Remote Pilot License

Kenya Civil Aviation Authority

Issued January 2025

Conferences & Workshops

Delegate – GDH Partner Days on Challenge Driven Education [March 2026]

Royal Institute of Technology – Stockholm, Sweden

Presenter - A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters [February 2026]

11th International Congress on Information and Communication Technology - London, United Kingdom

Delegate – Microsoft Faculty Skilling and Immersion Program (Cohort 3 – AI-Oriented Education) [March - June 2025]

Microsoft ADC & Microsoft Leap - Nairobi, Kenya

Delegate - 5th International Conference on Smart Sensors and Application [September 2024]

Universiti Teknologi Malaysia - Penang, Malaysia

Delegate – CMU Innovation Summit 2024 [January 2024]

Carnegie Mellon University Africa - Kigali, Rwanda

Organiser & Presenter – Fundamentals of Raspberry Pi 400 [November 2023]

Strathmore University - Nairobi, Kenya